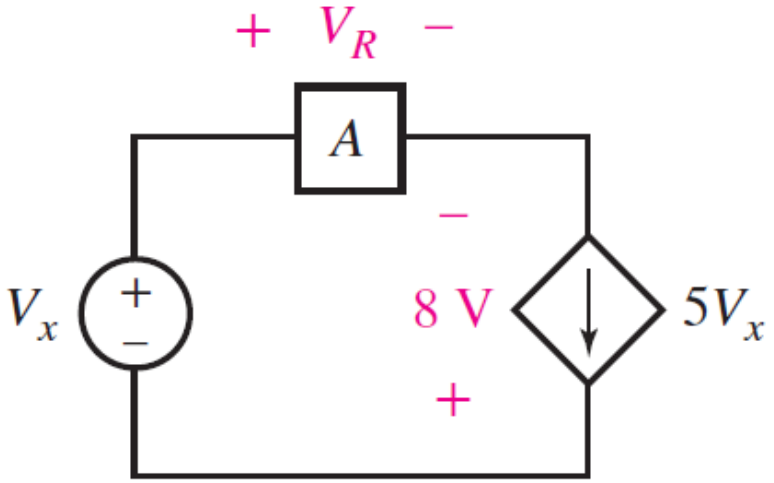


(a)For $V_R = 100$ V and $V_X = 92$ V, determine the power supplied by each element.

Current through dependent source is $I = (5)(92) = 460$ A

Power absorbed for Vx source: P= -(92V)(460A)=- 42.32kW

Power absorbed for element A: P= (100V)(460A)=46kW

Power absorbed for 5Vx dependent source: P= - (8V)(460A)= -3.68kW

(b) -42.32 kW - 3.68 kW +46 kW = 0

$$p = vi = \frac{v^2}{R}$$

R= 100 Ω, p=v^2/100 ($-2 \leq v \leq +2$)

